Artificial Intelligence — Lab 2 Report

1. PEAS Specifications

Delivery robot: Performance—successful delivery, collision avoidance, time, energy; Environment—corridors, doors/elevators, people, packages; Actuators—drive, steering/braking, gripper, alerts; Sensors—camera/depth, lidar/ultrasonic, IMU/encoders, package/location markers. LLM student-support agent: Performance—accuracy, usefulness, grounding, privacy, escalation; Environment—student messages, approved course knowledge, conversation, calendar/tool results; Actuators—text, links, authorized scheduling/escalation; Sensors—messages, context, retrieved documents, approved tool results. Main trade-off: usefulness versus safety/privacy.

2. Environment Classification

Both real systems are classified as partially observable and stochastic under the stated assumptions. The robot cannot observe every pedestrian intention or hidden obstacle, and physical outcomes can vary. The student-support agent cannot observe unstated student circumstances and student/tool responses are uncertain. A simplified simulator can be fully observable and deterministic if it exposes complete state and uses fixed transitions.

3. Stateful Controller

Temperature <18°C -> HEAT; >24°C -> COOL; otherwise OFF. Occupancy=0 has priority -> OFF. Previous target mode is stored. Same target -> NO-COMMAND; changed target -> SEND.

4. Trace

1 None, 20°C/1 -> OFF/SEND; 2 OFF, 20°C/1 -> OFF/NO-COMMAND; 3 OFF, 18°C/1 -> OFF/NO-COMMAND; 4 OFF, 24°C/1 -> OFF/NO-COMMAND; 5 OFF, 17°C/1 -> HEAT/SEND; 6 HEAT, 25°C/1 -> COOL/SEND; 7 COOL, 25°C/0 -> OFF/SEND.

5. Edge Case

Exact 18°C and 24°C produce OFF because the rules use strict < and > comparisons. An implementation using <= or >= would change the boundary behavior. Boundary tests make this visible.

6. Responsible Practice

Robots should prioritize safety under uncertainty. Student-support agents should use least-privilege tools, protect privacy, state uncertainty, avoid inventing policies, and escalate consequential cases.

References

A. M. Turing (1950), 'Computing Machinery and Intelligence,' Mind, 59(236), 433–460. https://doi.org/10.1093/mind/LIX.236.433

S. Russell and P. Norvig (2021), Artificial Intelligence: A Modern Approach, 4th ed., Pearson.

V. Vapnik (1998), Statistical Learning Theory, Wiley. (Background on uncertainty/generalization; used only as supporting conceptual context.)

Course-provided Lab 2 task specification and starter-rule assumptions. No external code was copied.